Akash Dubey

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AI / ML Engineering Experience

Lykke San Francisco, CA

Founding Software Engineer *Sep 2025 – Present*

- Engineered adaptive RAG pipelines across MongoDB, Zilliz/Milvus, S3, Gemini Files API, and SSE streaming, including source inference, citation events, token-budgeting, and fallback paths for extracted text vs. full-file reasoning.
- Built core AI workflows for an AI study platform used to ingest Canvas LMS courses, assignments, announcements, notes, and uploaded documents into class-scoped knowledge bases for contextual chat and study generation.
- Created a distributed course-wiki generation engine that crawls sources, synthesizes structured sections, embeds content for retrieval, and runs tenant-isolated background builds via SQS, MongoDB leases, Redis rate limiting, and AWS ECS.

Scarlet Sync New Brunswick, NJ

Founder *Jan 2025 – Present*

- Developed a production AI assistant with streaming tool-calling and grounded retrieval over schedules, degree plans, and profile context, enabling automated schedule building across Web, ChatGPT, Claude, and iOS.
- Architected a robust data ingestion pipeline using Python and LLMs to reverse-engineer legacy data, indexing over 2,500 classes and 500+ degree programs for real-time querying.
- Built a modern Rutgers academic planning platform with thousands of users for degree, schedule, and class planning as well as campus discovery, utilizing AI-assisted parsing of unstructured data from legacy university services.

New York Life Insurance New York, NY

Software Engineer Intern, TDAV Team *May – Aug 2026*

- Created an AI text and voice agent for processing insurance claims for clients and 12,000 Customer Service Representatives on AWS, with access to NYL services, knowledge, and forms.
- Built multi-agent systems hosted on NYL's Cloud to improve website development in AEM, Pega code, Tungsten document processing, and mainframe modernization; gained working experience extracting business rules from COBOL and low-code systems.
- Constructed a company knowledge base from 13,000 documents across SharePoint, Confluence, Jira, and the Intranet, unifying business logic for AI and human users and supporting cross-functional engineering work.

Samaritan Scout Cranford, NJ

Full Stack Engineer Intern *May 2023 – Aug 2025*

- Developed an agentic scraping system leveraging OpenAI and Gemini structured outputs to autonomously extract data from over 100,000 non-profit websites, reducing manual data entry time by over 90%.
- Built a full-stack semantic search engine (React, TypeScript, PostgreSQL) over 35,000+ volunteer opportunities with hallucination filters for accurate matching.

Selected AI Projects

Knowledge Base and AI Agent Workflow Manager | *MCP, Coding Agents, Hybrid Retrieval*

- Built KRAIL (pypi), a local-first, repo-backed knowledge runtime that structures raw notes into ontology-aware topic pages, source ledgers, claim records, and verification runs for evidence-grounded agent memory.
- Implemented MCP/CLI and agent infrastructure for auditable coding-agent workflows (software factories, auto-research, company brain), including runner dispatch, work orders, session logs, workflow validation, and hybrid keyword-graph-vector retrieval.

Local Coworker: Native macOS Computer-Use Agent | *Swift, MCP, Gemini Vision* (private code)

- Built a native desktop agent for autonomous computer use, integrating Gemini Vision with macOS Accessibility APIs, AppleScript, and JXA, and using MCP to orchestrate specialized sub-agents (UI Interaction, Shell, App Research) across complex multi-app workflows.

SLM Optimization & Post-Training (GSM8k) | *PyTorch, HPC*

- Fine-tuned small language models (SLMs) on the GSM8k math reasoning dataset on the Rutgers HPC (Amarel) cluster, using quantization and structural pruning to maximize inference throughput on constrained compute.

Education

Rutgers University Honors College, School of Arts and Sciences — New Brunswick, NJ

Bachelor of Science in Computer Science and Mathematics — *Anticipated May 2028*

- **GPA**: 3.95/4.0 | SAT: 1580 | Dean's List. **Coursework**: Systems Programming, Stochastic Processes, Computer Architecture, Algorithms, Data Structures, Math Theory of Probability, Honors Calculus III & IV.

Research

Rutgers Economics Labs — New Brunswick, NJ

President (formerly Research Director, Team Lead) — *Oct 2024 – Present*

- Architected RAIL (Rutgers Agentic Intelligence Labs), an automated research pipeline to collect, organize, analyze, and synthesize economic data using data ontologies and coding-agent workflows; lead quantitative research for partners including NJDOL, NJDEP, Federal HUD, and the U.S. Census Bureau.

Rutgers CS & Cancer Institute (Algorithmic Robotics Lab; Kwan Lab) — New Brunswick, NJ

Research Assistant, ML Systems & Computational Biology — *Jan 2026 – Present*

- Implemented CUDA versions of pRRTC and FCIT* for high-dimensional robot motion planning; developing a graph-based search policy (GaP) that learns from planner traces to rank symbolic action skeletons.
- Built deep-learning genomics pipelines (AlphaGenome in silico deletion screens, DeepSTARR enhancer inference) and a 3D chromatin polymer simulator, validated with 70+ automated tests on HPC.

Awards & Skills

Skills: RAG, Vector DBs (Zilliz/Milvus), Multi-Agent Orchestration, MCP, LLM Tool-Calling, SSE Streaming, Evals & Hallucination Filters, Fine-Tuning, Quantization, Retrieval
Providers & Frameworks: OpenAI, Anthropic (Claude), Gemini, PyTorch, Transformers, Generative AI
Languages & Infra: Python, C++, CUDA, TypeScript, SQL, Java, R, Bash, COBOL (working knowledge), Git; AWS (ECS, SQS, S3), MongoDB, Redis, PostgreSQL, HPC
Awards: 8090 AI Top Coder (5th), Rutgers Shark Tank (2nd and 3rd), ASA Fall Data Challenge (Top 3), NSF I-Corps.